Information Retrieval
-Introduction-

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Faculty of Computer Science
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Semarang

Muljono

Riwayat Pendidikan :

- SD Kanisius Kobong II Semarang, lulus tahun 1983
- SMP Kanisius Raden Patah Semarang, lulus tahun1986
- SMA Negeri 3 Semarang, lulus tahun 1989
- S1 Matematika UNDIP Semarang, lulus tahun 1996
- S2 Teknik Informatika STTIBI Jakarta, lulus tahun 2001
- Internship Program Doktoral Speech Processing- Tokyo University of Technology (TUT), Japan, lulus tahun 2014
- S3 Teknik Elektro, ITS Surabaya, lulus tahun 2016

Pengalaman Tambahan :

- Workshop Datamining, PENS ITS Surabaya, 2006
- Soft Computing and Pattern Recognition (SOCPAR), Universiti of Teknikal Malaysia Melaka (UTeM), Malaysia, 2009.
- Seminar dan Workshop Soft Computing, South China University of Technology (SCUT), Guangzhou, China, 2013
- Visiting Professor (Dosen Tamu) di University of Split Croatia, Program Beasiswa Erasmus^{+} Uni Eropa, 2018
- Workshop Big Data Analaysis dan IOT, Big Data Alibaba, Fozhou Polythechnic, Fouzhou China, November sd Desember 2019

Topics

1. Introduction
2. IR Model
3. Document Classification: k-NN and Naive Bayes
4. Document Clustering: K-Means
5. Document Clustering: Hierarchical Agglomerative Clustering (HAC)
6. Document Clustering: Self Organizing Map (SOM)
7. Features Reduction
8. Document Summarization
9. Rapidminer
10. Weka
11. Lucene
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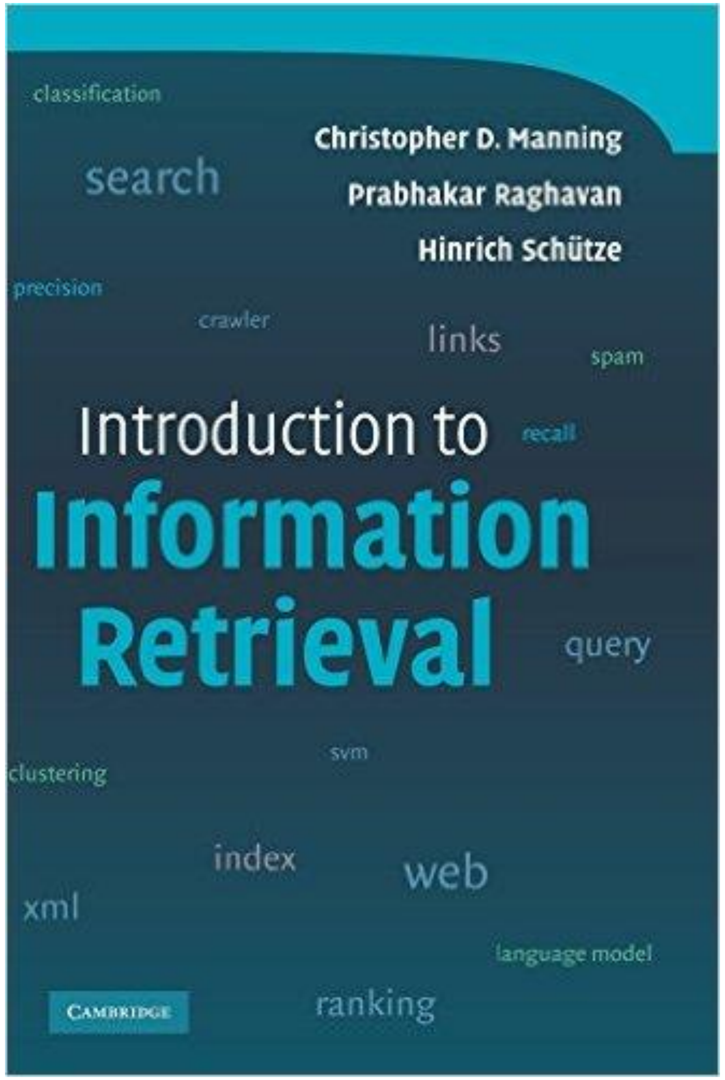
Handbook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Paper

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- Annually
- 2-6 pages
- IEEE, presentation is compulsory

Journal

- 2 times or more publication every year
- More than 6 pages

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Clustering results of image searches by annotations and visual features

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A R T I C L E I N F O

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Category model
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Image annotation

A B S T R A C T

Image on web has become one of the most important information for browsers; however, the large number of results retrieved from images search engine increases the difficulty in finding the intended images. Image search result clustering (ISRC) is a solution to this problem. Currently, the ISRC-based methods separately utilized textual and visual features to present clustering result. In this paper, we proposed a new ISRC method as called Incremental-Annotations-based image search with clustering (IAISC), which adopted annotation as vxtual features and category model as visual features. IAISC can provide clustering result based on the semantic meaning and visual trail; further, presented by the iteratively structure, a user can obtain the intended image easily. The experimental result shows our method has high precision that the average precision rate is 73.4%; particularly, the precision rate is 96.5% when the user drills down the intended images till the last round. Regarding efficiency, our system is one and a half times as efficient as the previous studies.

1. Introduction

Image on web has become one of the most important information for users these days. Since there are a huge number of images on the web, it is hard to find the intended images by simple query. The results of a query may contain several topics; for example, a query "Pluto" in Google Images, the results contain two different scope of semantic meaning: one is Pluto in the solar system; the other is the dog named "Pluto" in Disney world (Cai et al., 2004). It is worth noting that the results of the two scopes are mingled, that means it is hard to identify the required images immediately. Since the search engine does not clustering the results beforehand, the users have to browse all the images through page by page, and then they should manually organize these images.

Recently, web image search engine, like Google image, only returns a small fixed number of images (Wang et al., 2007). In the perspective of users, it becomes even harder to find the intended images when different scopes mingled together. To solve the problem, technique of image search result clustering (ISRC) was proposed (Cai et al., 2004; Wang et al., 2007; Ding et al., 2008; Jing et al., 2006) which is clustering images based on visual or textual features.

Regarding the visual features, such as color or texture, can be used to find similar images (Cai et al., 2004); however, the features have some limitations that it is hard to present the semantic meaning (called semantic gap) with the visual features and it takes too much time to extract the visual features from high-definitional and high-dimensional images.

Regarding the text features, it means the features of the text surrounding the image or other text sources. Text-based ISRC (Wang et al., 2007; Ding et al., 2008; Jing et al., 2006) extracted key phrases from the texts to clustering images into several clusters in a reasonable time, compared to the time to clustering images by visual features. With the text features, images in each cluster then have semantic meaning and can be used for further application. Although the images are annotated with

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Journal

Grouping WWW Image Search Results by Novel Inhomogeneous Clustering Method

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Abstract

In this paper, a novel inhomogeneous clustering method is proposed for grouping web images. It is used to re-organize the search result of web image search engines into a hierarchical structure so that the users can conveniently browse the search result. This method takes into account various features associated with web images, and treats them in different ways. For the surrounding text extracted from the containing web pages, co-clustering approach is adopted; for low-level features of the image content and other features, one-way clustering approach is adopted. The clustering results of different approaches are combined together to produce the final image groups. Experimental results demonstrate the effectiveness of the proposed method.

1. Introduction

WWW image search engines [1] are powerful tools to search for digital images on the Internet by keywords. Unlike traditional image databases with manually labeled annotations, web image search engines index images with some text-based features, such as image file names or the surrounding text in containing web pages. Those features may be regarded as an approximate description of the image content. When the user submits a keyword query, the system typically produces a ranked list of images according to the relevance of image's text description to the user's query.

However, because of the ambiguities of keywords, the results of the existing search engines are still not satisfactory in many cases. Even if all images returned are relevant to the input keywords, it is yet difficult for the user to find the right images with his/her intended concepts or visual styles. Obviously, even if different users use the same keywords to search images, their objectives may be different. The one-dimensional search results produced by current search engines can not meet requirements of different users. Therefore, it will be quite useful if we can automatically group search results into different clusters in terms of concepts and visual styles. In this manner, users are allowed to view the search results through a few clusters rather than jumbled images. Some studies also show that grouping images by visual features can help the user browse search results [2].

As web images are indexed with text information, the co-clustering method [5] used to cluster both terms and documents may be adopted here. However, as the surrounding text automatically extracted from containing web pages are not accurate enough, other information of images such as low-level features and hyperlink structures, should be taken in account as well. But those features are quite different in nature with each other: discrete or continuous, dense or sparse, high-dimensional or low-dimensional. How to use them simultaneously is a challenging problem.

In this paper, we propose a novel parallel, hybrid clustering algorithm to process inhomogeneous information naturally. Every feature can select its "favorite" clustering algorithm, and its "contribution" can be merged into a global loss function. Using this algorithm, we can cluster search results of our web image search engine by keywords, low-level features and hyperlink information, and got encouraging experimental results.

The paper is organized as follows. Firstly, in Section 2, we introduce the related works. The detailed explanation of the proposed algorithm is presented in Section 3, including the flowchart of our algorithm and the discussion of the convergence. The experimental results are given in Section 4. Concluding remarks appear in Section 5.

2. Related Work

In general, existing clustering algorithms may be classified into two types: one-side clustering and parallel clustering. The one-side clustering, also named one-way clustering, clusters along one dimension based on similarities with respect to other dimension (e.g. image

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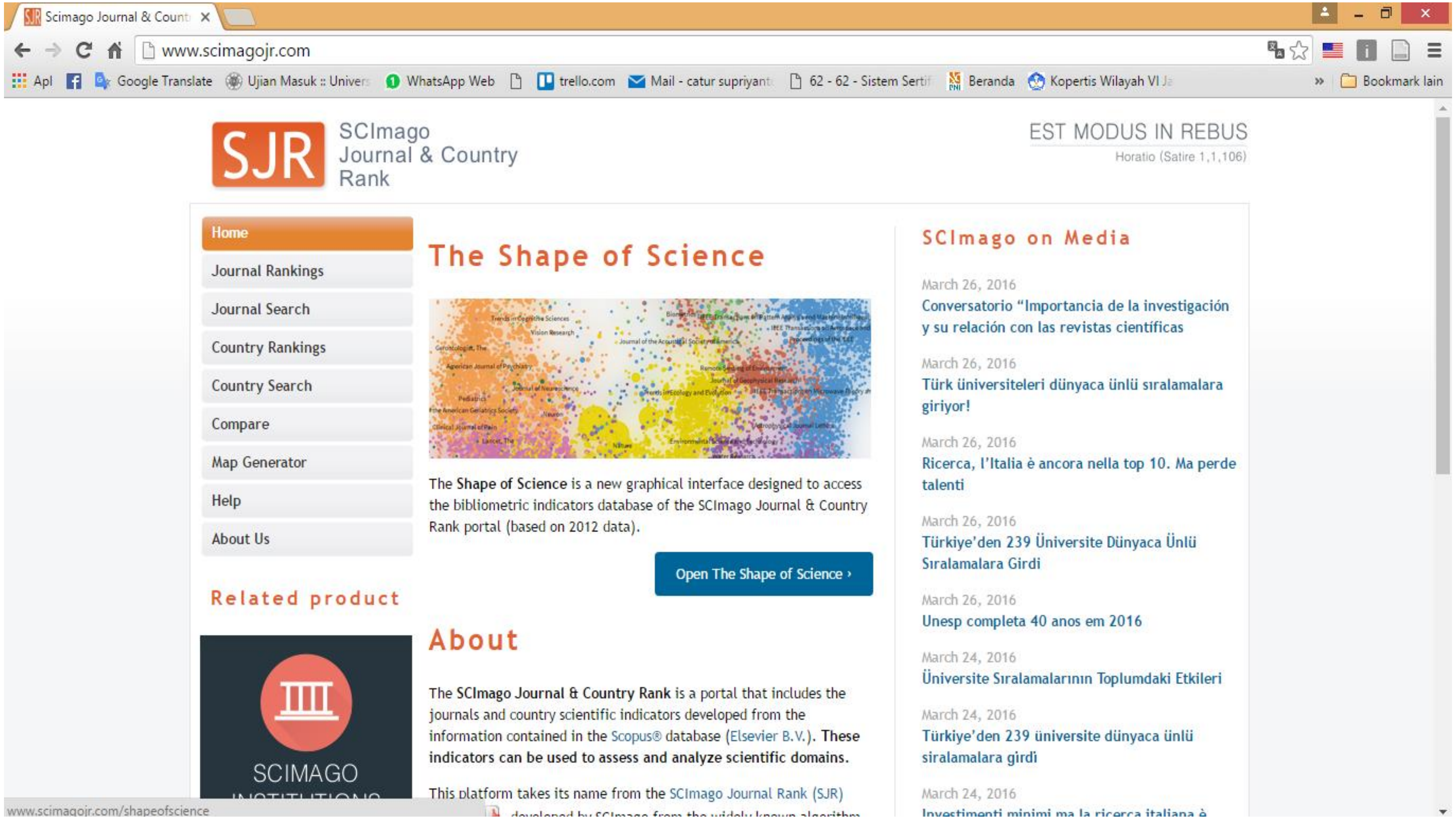
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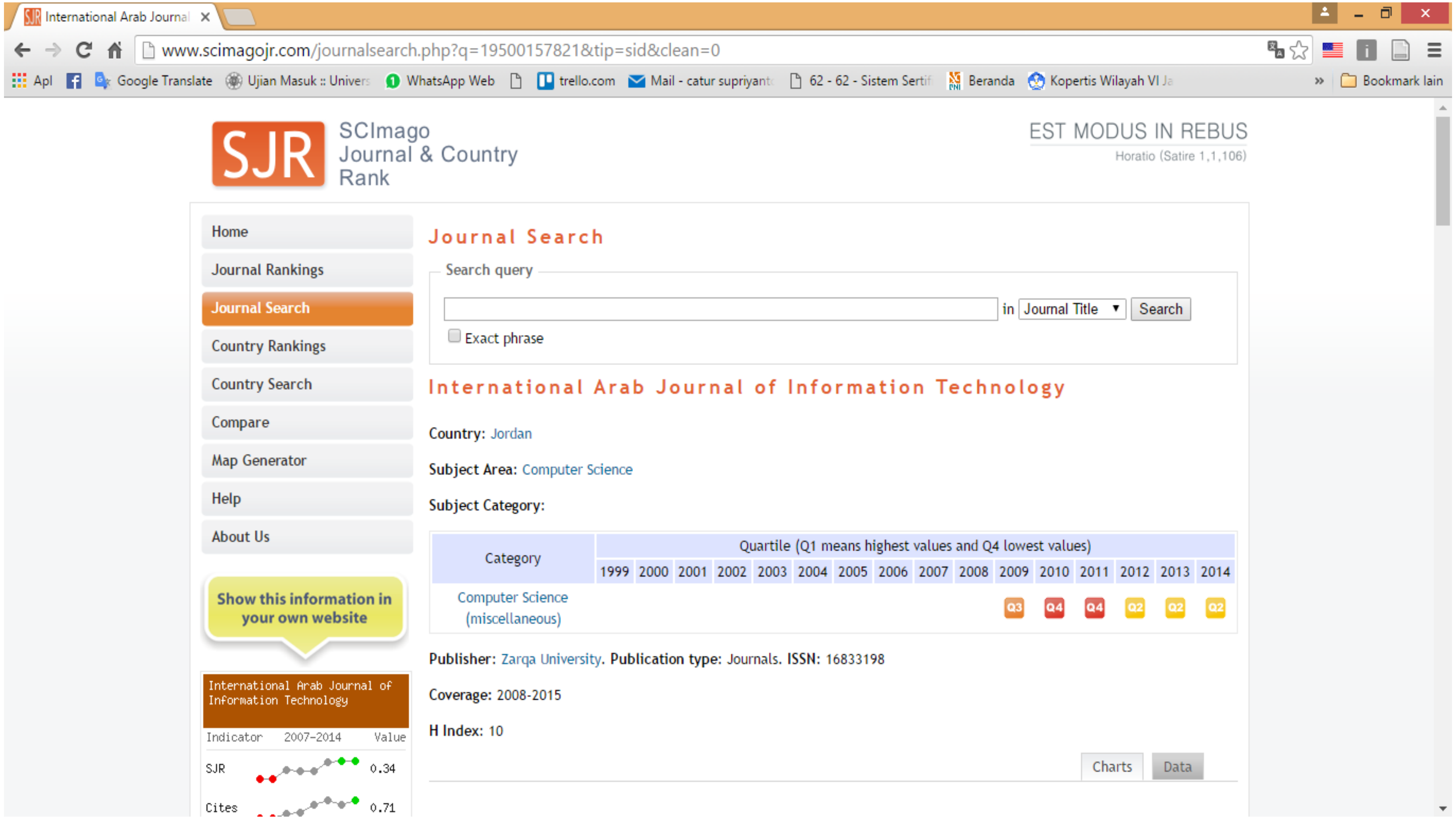

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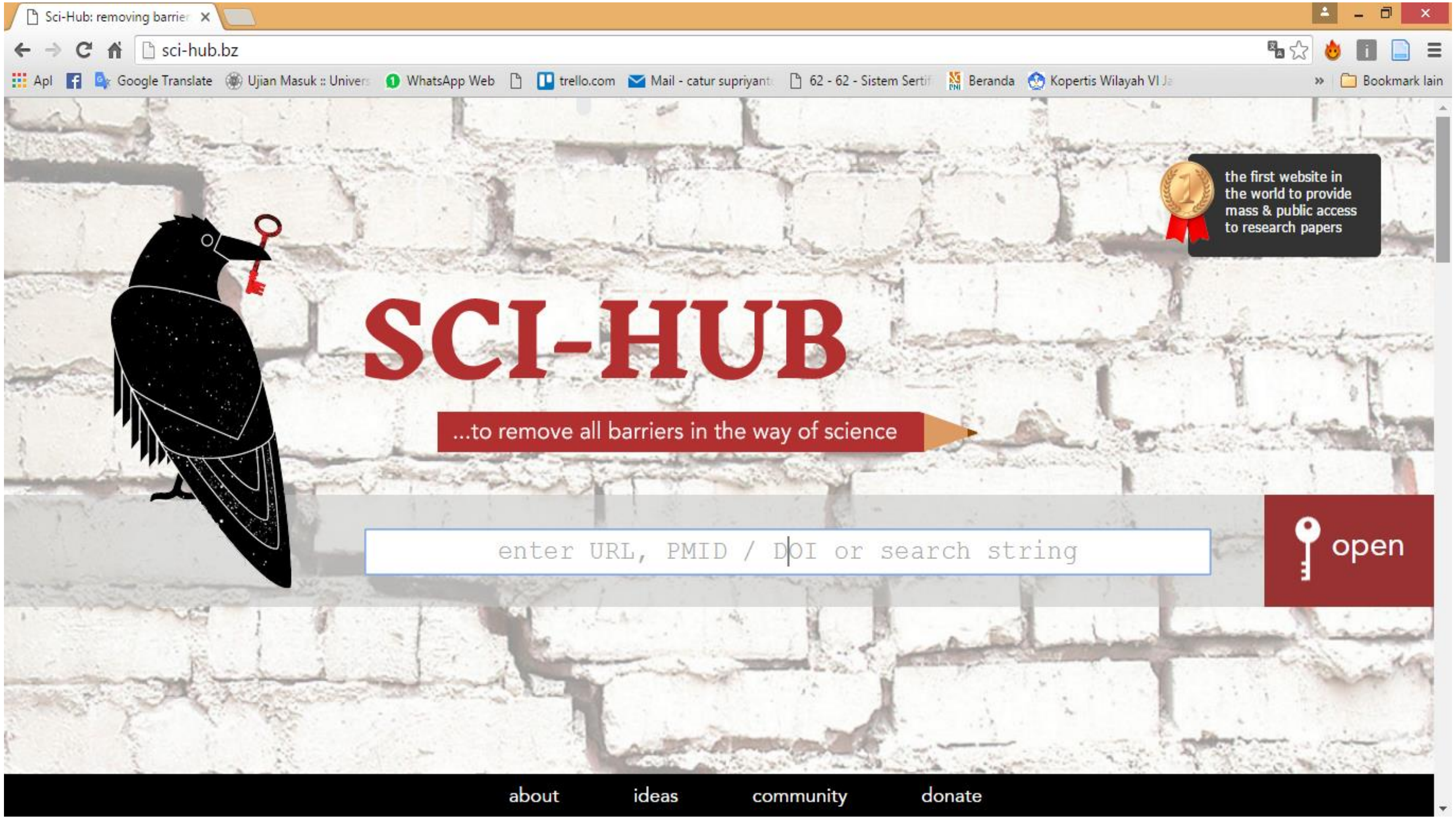

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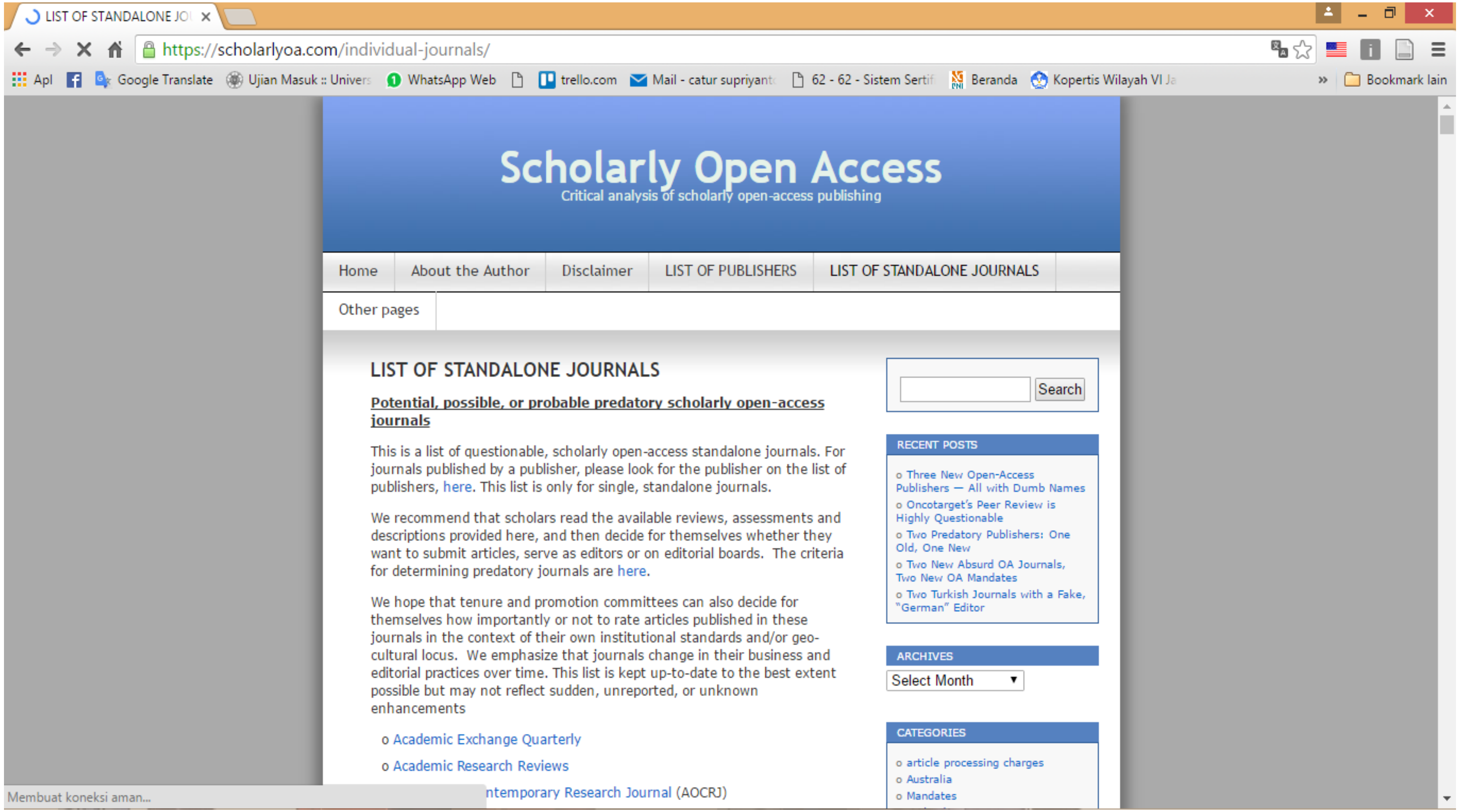

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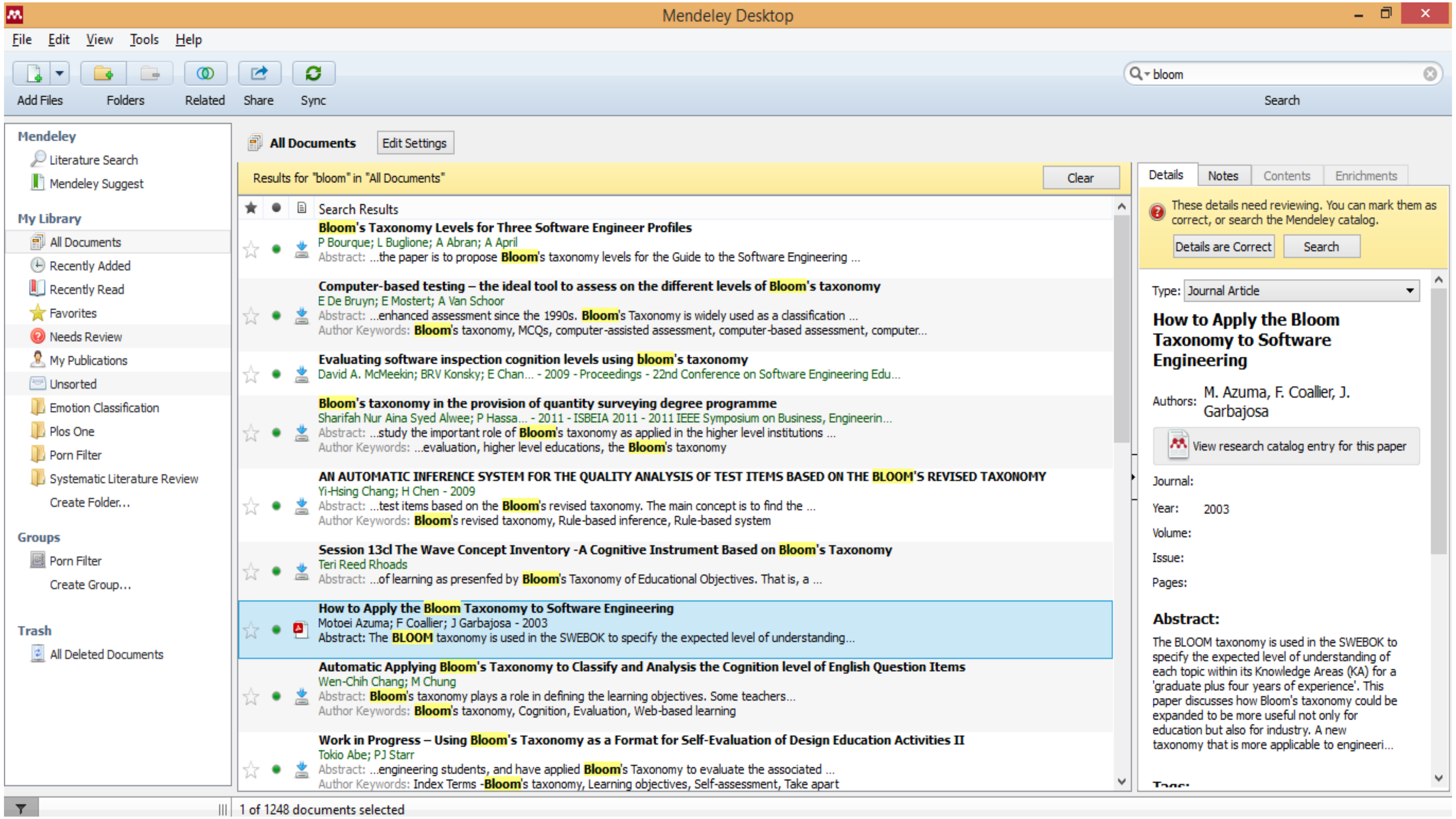

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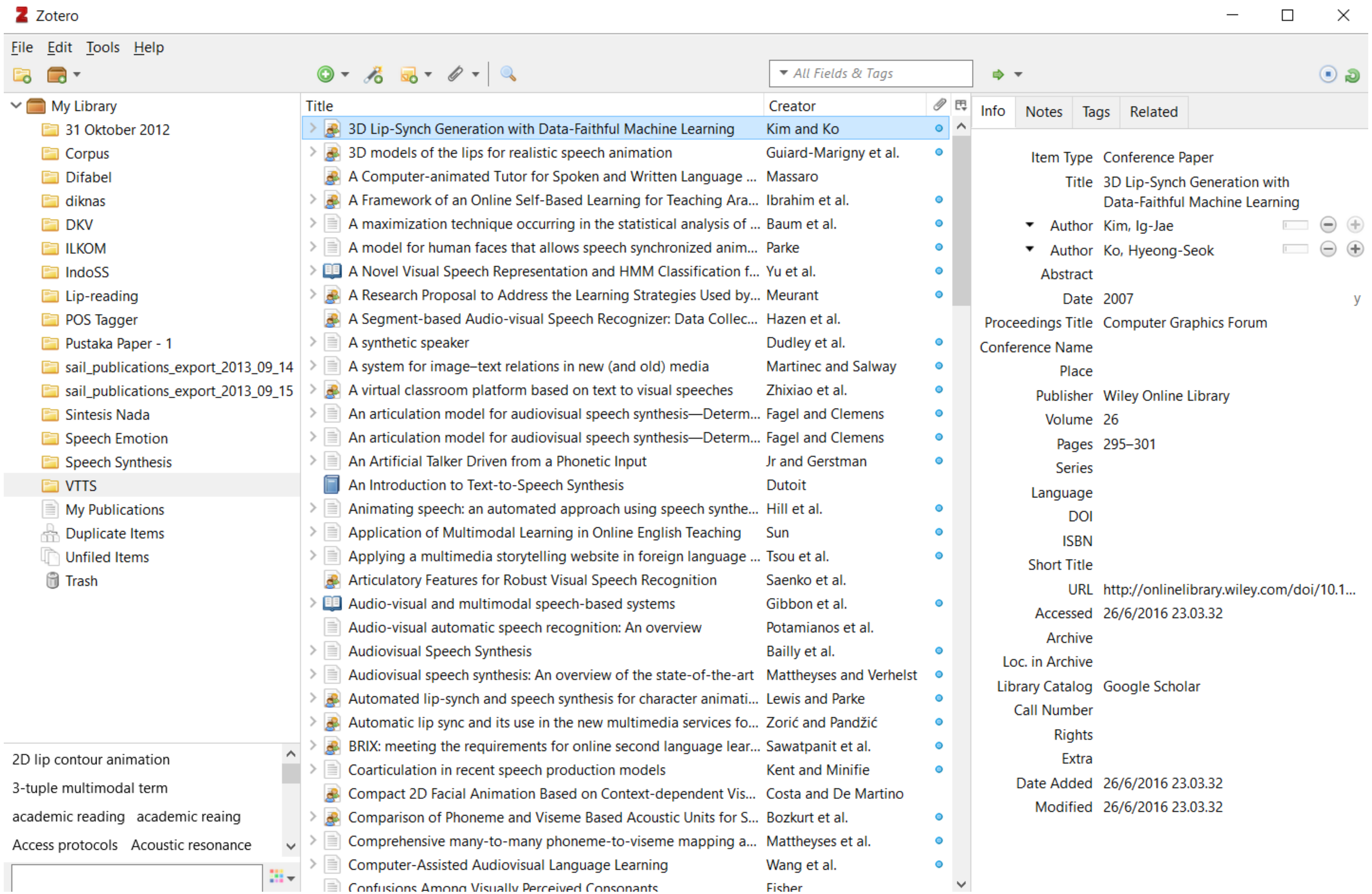

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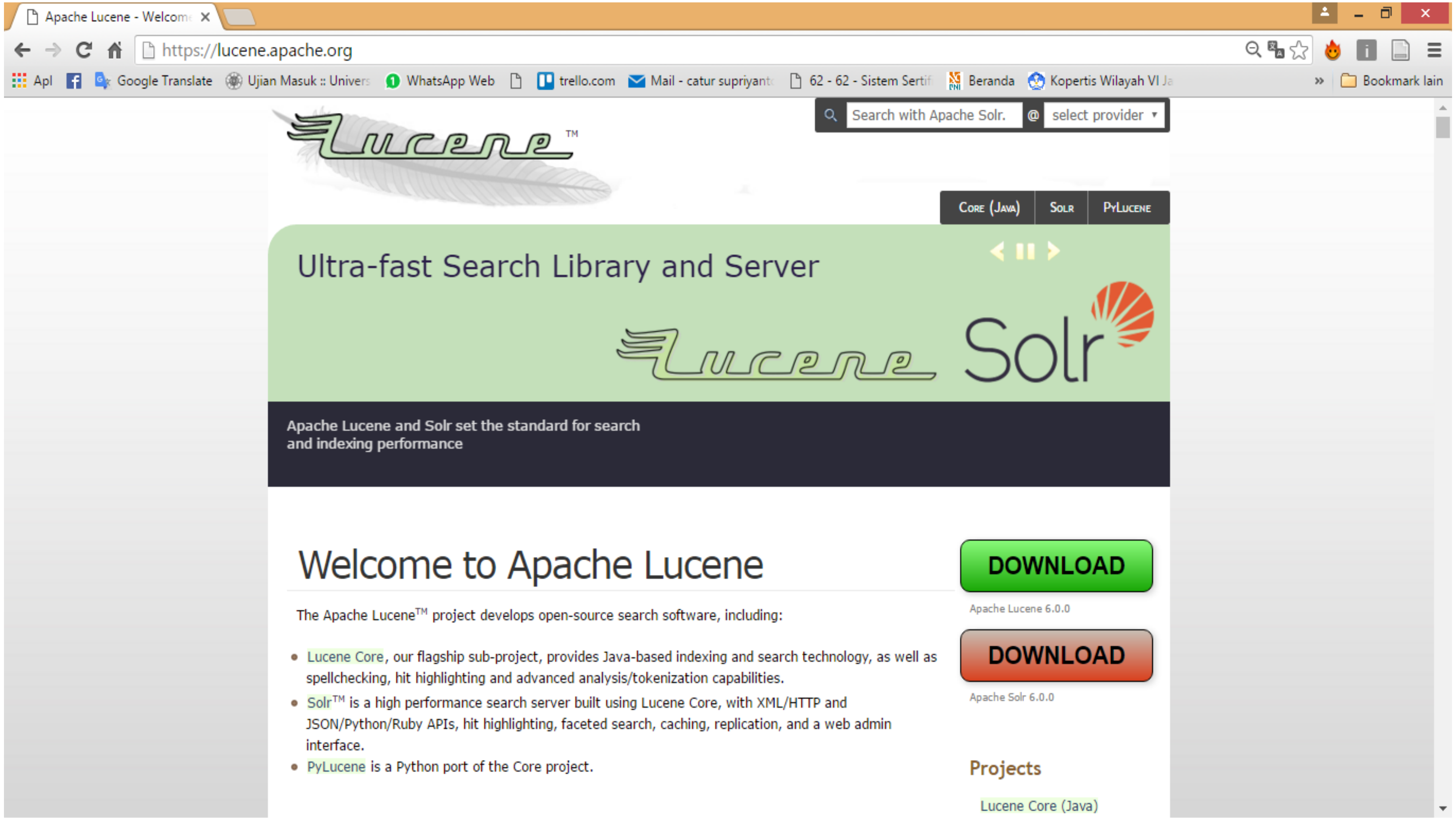

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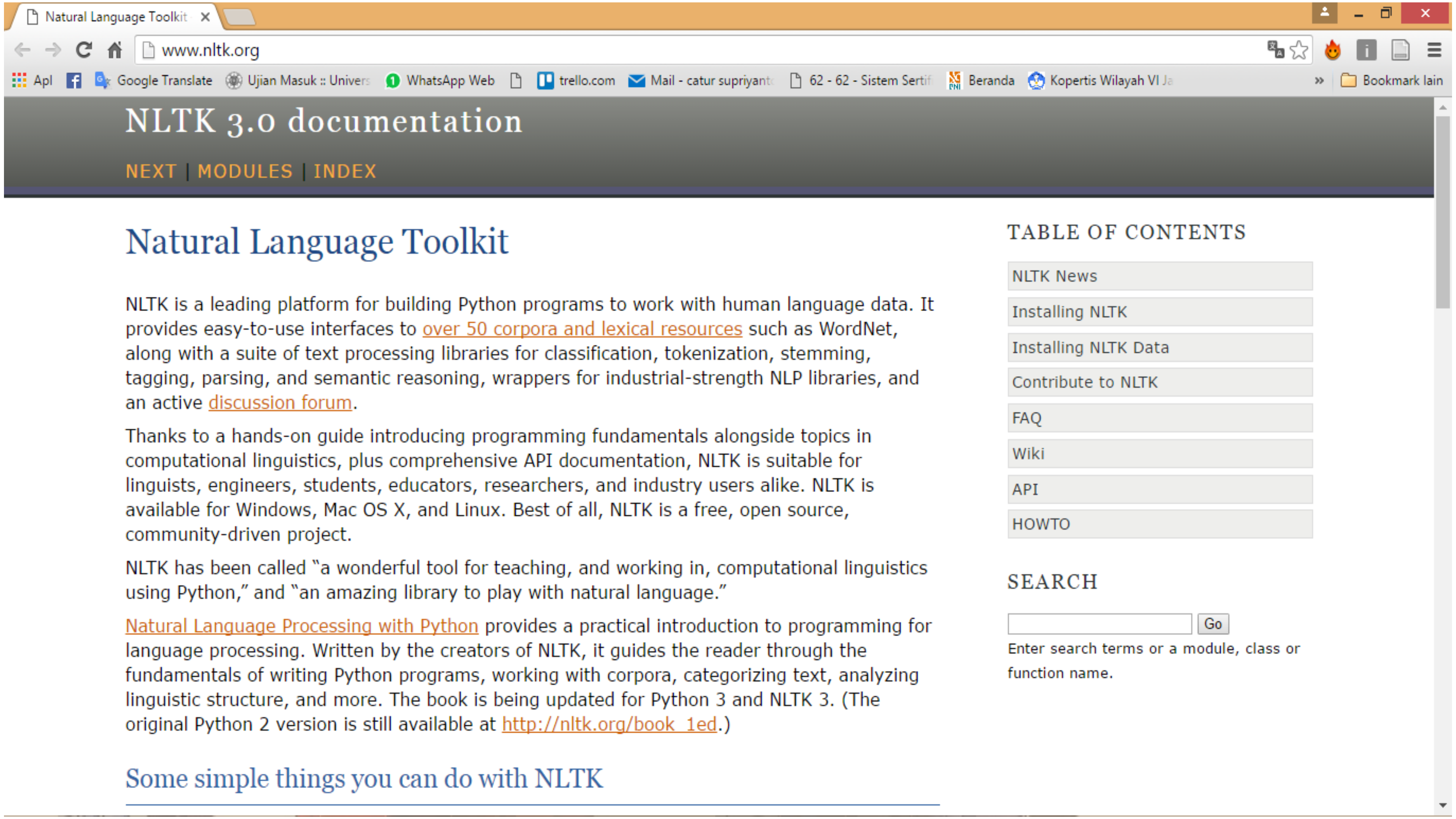
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Natural Language Toolkit
NLTK is a leading platform for building Python programs to work with human language data. It provides easy-to-use interfaces to over 50 corpora and lexical resources such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries, and an active discussion forum.
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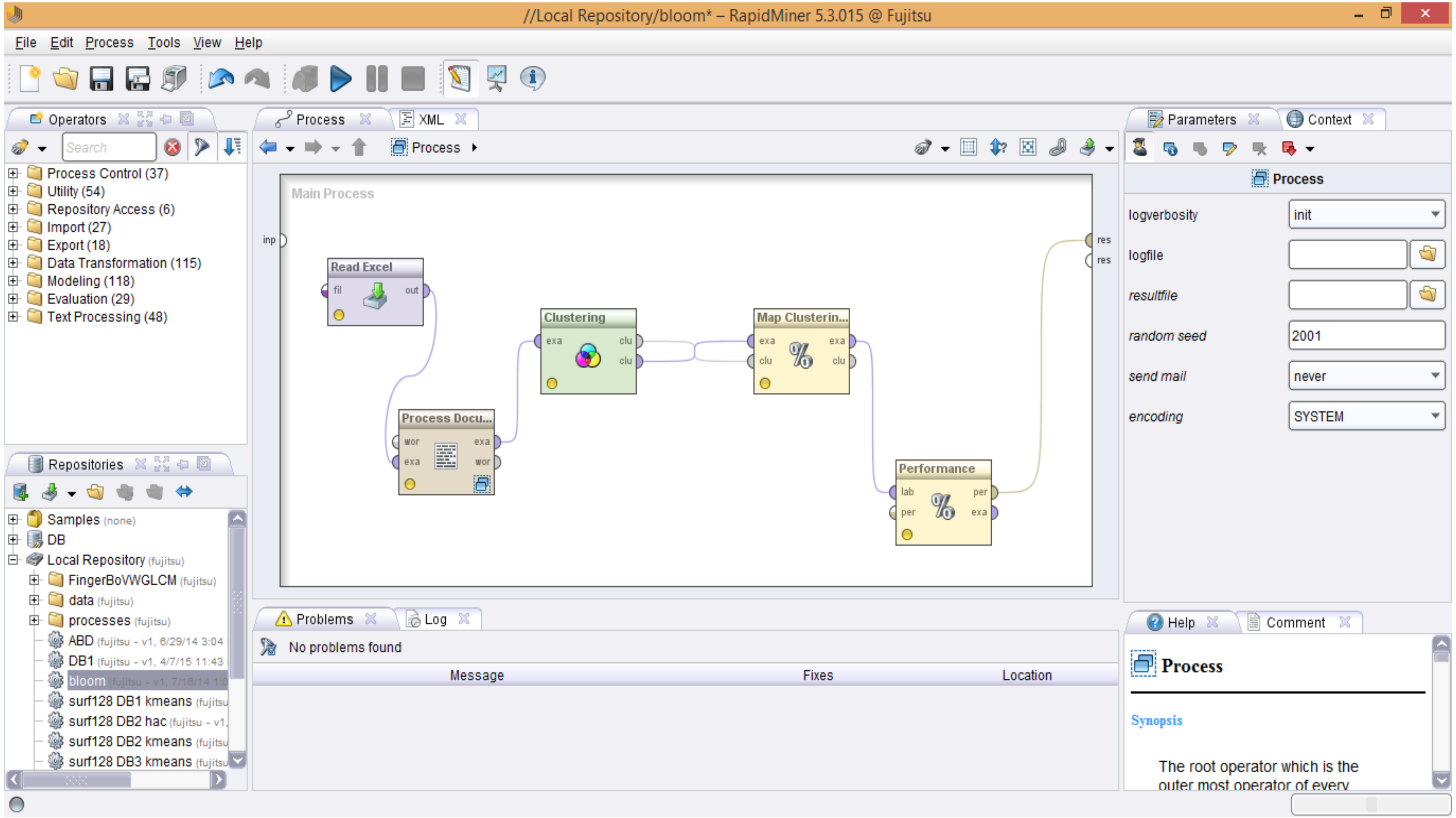
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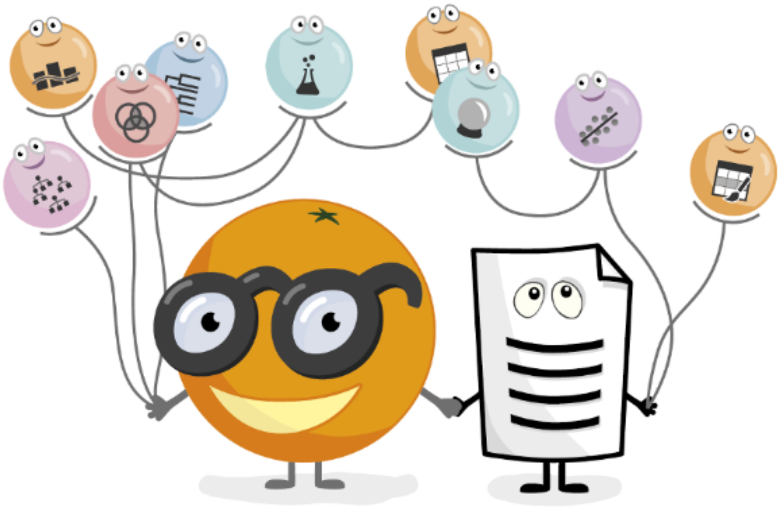

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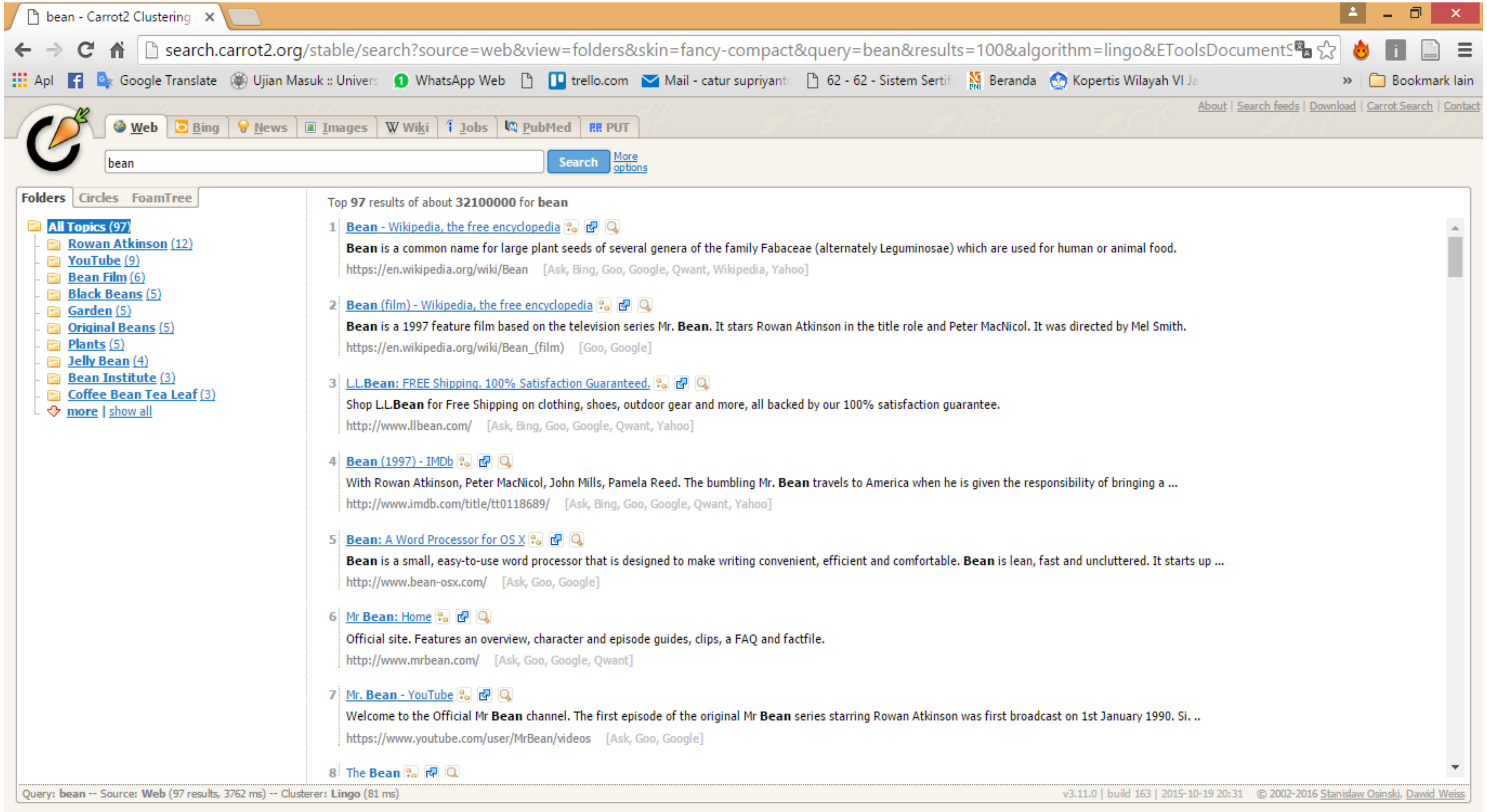

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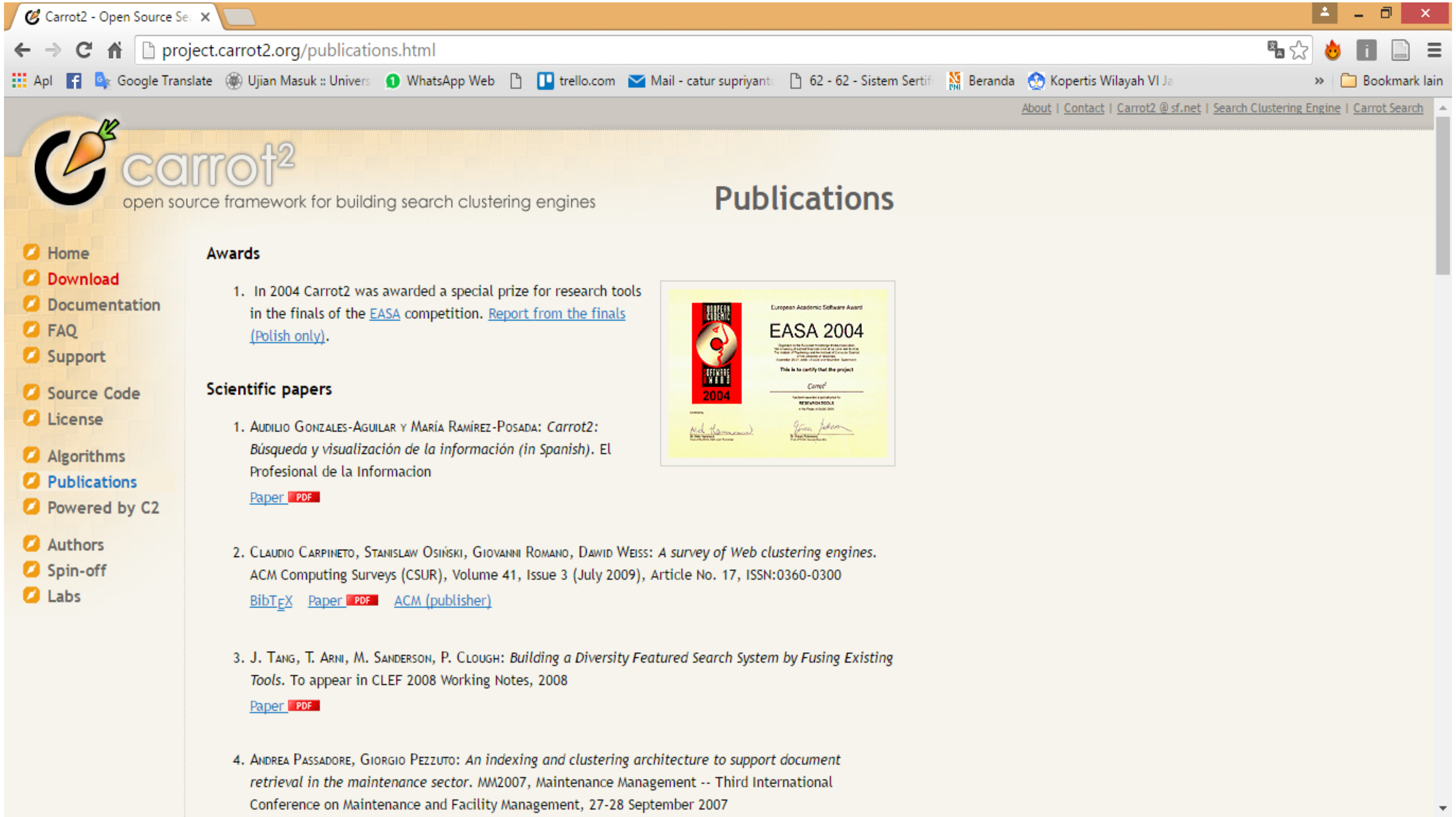

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1. In 2004 Carrot2 was awarded a special prize for research tools in the finals of the EASA competition. Report from the finals (Polish only).
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Scientific papers
1. Audilio Gonzales-Aguilar y María Ramírez-Posada: Carrot2: Búsqueda y visualización de la información (in Spanish). El Profesional de la Informacion
Paper PDF
2. Claudio Carpineto, Stanisław Osiński, Giovanni Romano, Dawid Weiss: A survey of Web clustering engines. ACM Computing Surveys (CSUR), Volume 41, Issue 3 (July 2009), Article No. 17, ISSN:0360-0300
BibTEX Paper PDF ACM (publisher)
3. J. Tang, T. Arni, M. Sanderson, P. Clough: Building a Diversity Featured Search System by Fusing Existing Tools. To appear in CLEF 2008 Working Notes, 2008
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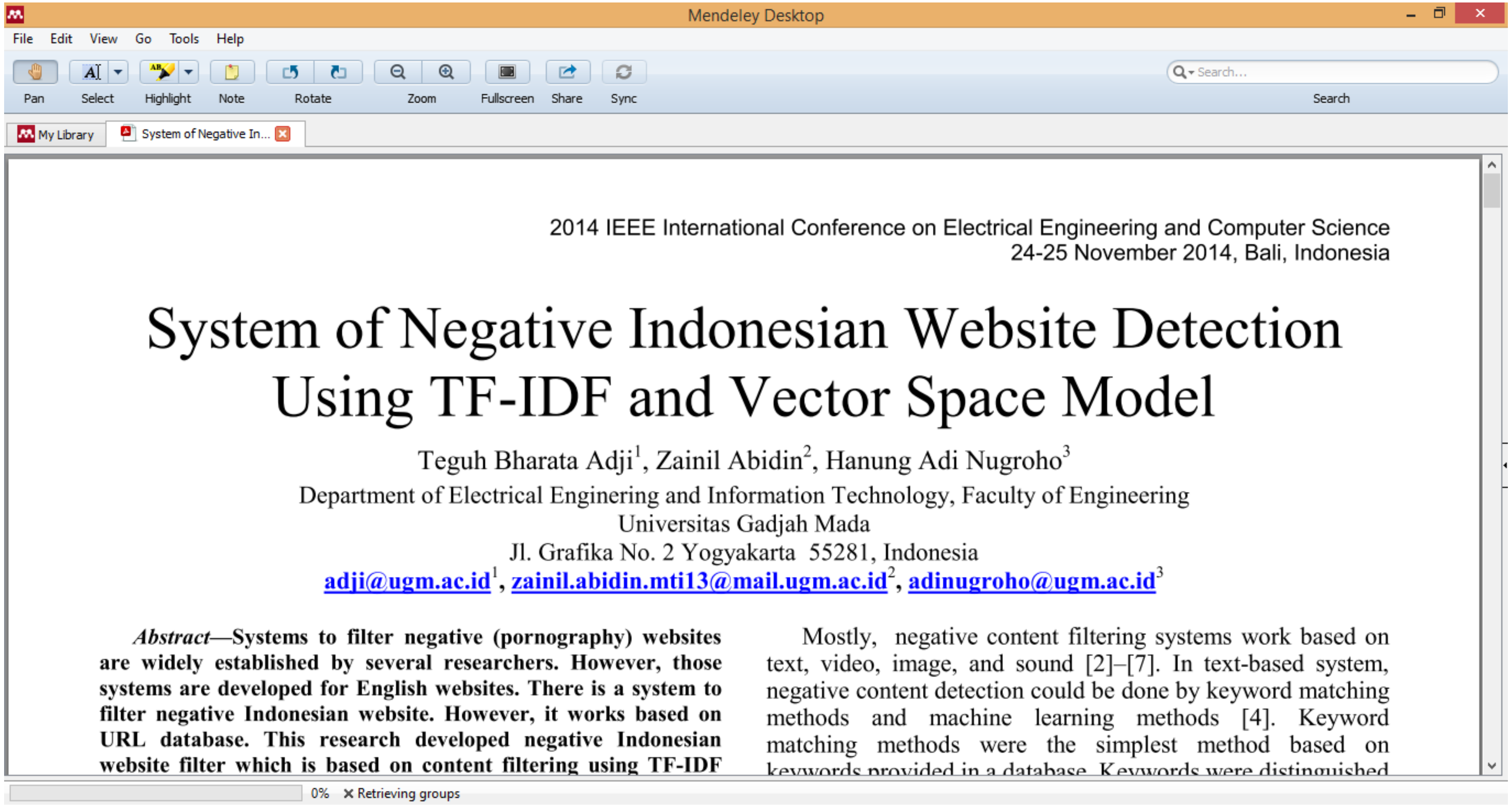
4. Andrea Passadore, Giorgio Pezzuto: An indexing and clustering architecture to support document retrieval in the maintenance sector. MM2007, Maintenance Management -- Third International Conference on Maintenance and Facility Management, 27-28 September 2007

Porn Detection



2014 IEEE International Conference on Electrical Engineering and Computer Science
24-25 November 2014, Bali, Indonesia

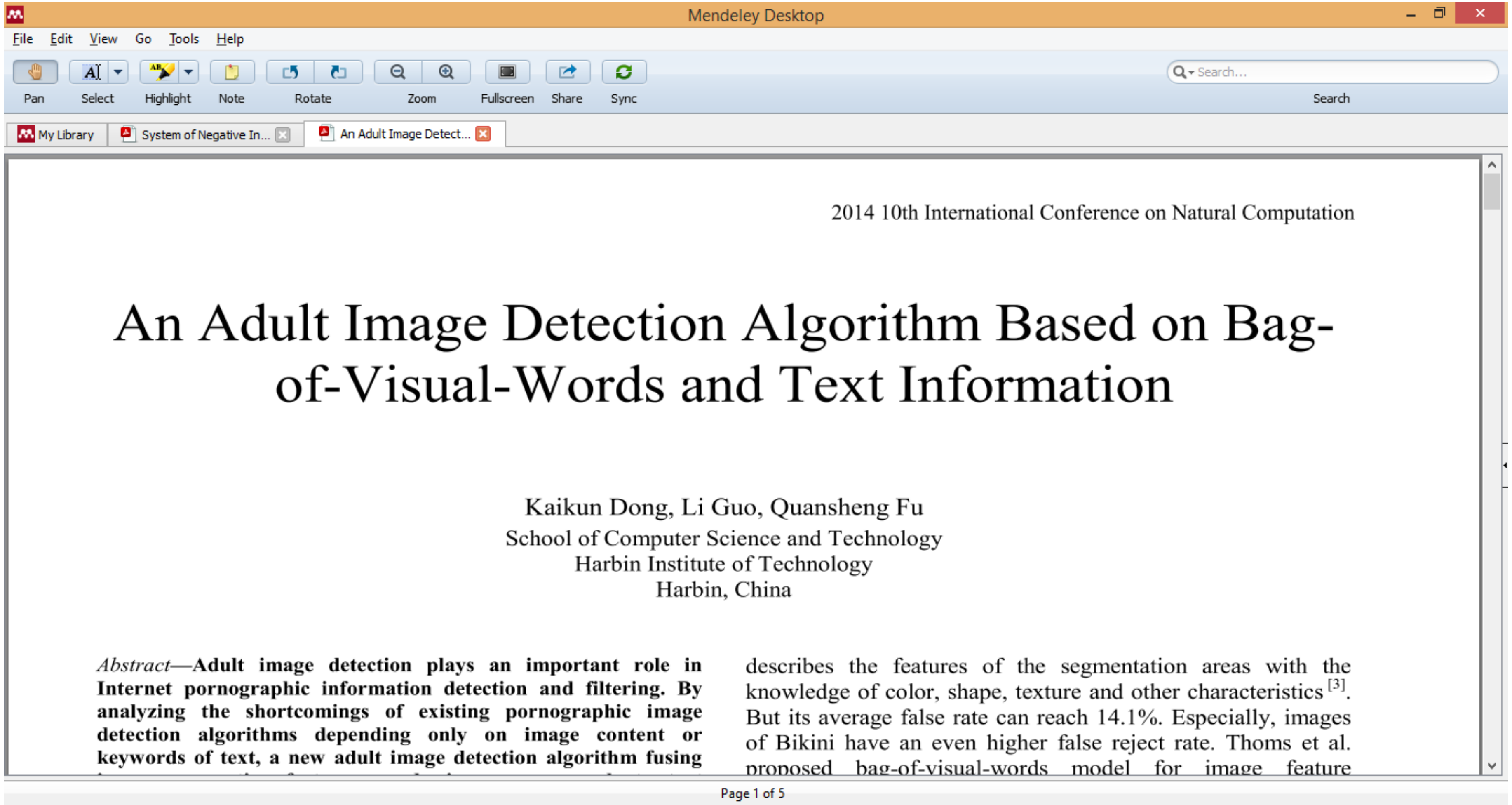
System of Negative Indonesian Website Detection Using TF-IDF and Vector Space Model

Teguh Bharata Adji[1], Zainil Abidin[2], Hanung Adi Nugroho[3]
Department of Electrical Enginering and Information Technology, Faculty of Engineering
Universitas Gadjah Mada
Jl. Grafika No. 2 Yogyakarta 55281, Indonesia
adji@ugm.ac.id[1], zainil.abidin.mti13@mail.ugm.ac.id[2], adinugroho@ugm.ac.id[3]

***Abstract*—Systems to filter negative (pornography) websites are widely established by several researchers. However, those systems are developed for English websites. There is a system to filter negative Indonesian website. However, it works based on URL database. This research developed negative Indonesian website filter which is based on content filtering using TF-IDF**

Mostly, negative content filtering systems work based on text, video, image, and sound [2]–[7]. In text-based system, negative content detection could be done by keyword matching methods and machine learning methods [4]. Keyword matching methods were the simplest method based on keywords provided in a database. Keywords were distinguished

Text and Image Based Porn Detection



2014 10th International Conference on Natural Computation

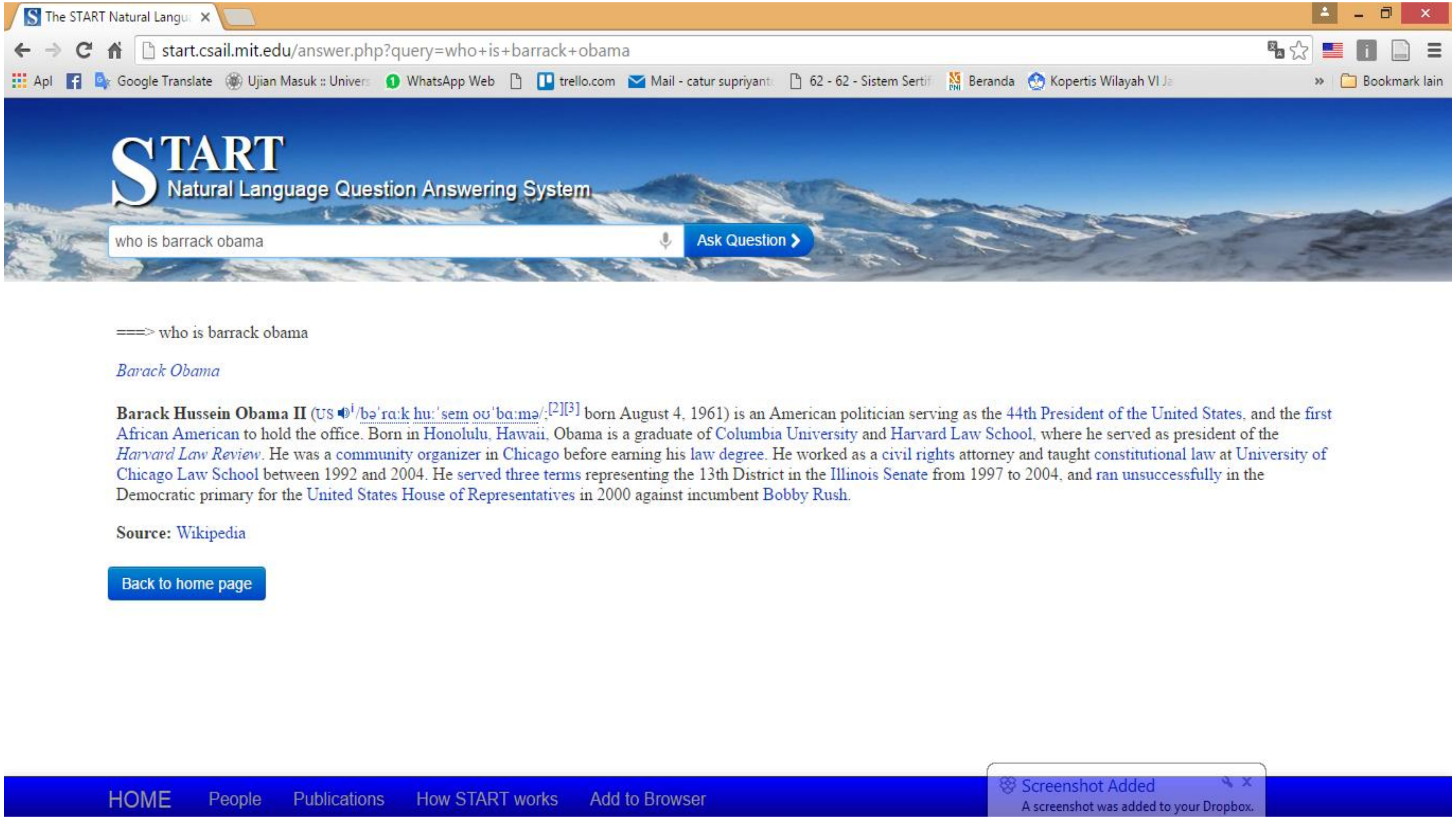
An Adult Image Detection Algorithm Based on Bag-of-Visual-Words and Text Information

Kaikun Dong, Li Guo, Quansheng Fu
School of Computer Science and Technology
Harbin Institute of Technology
Harbin, China

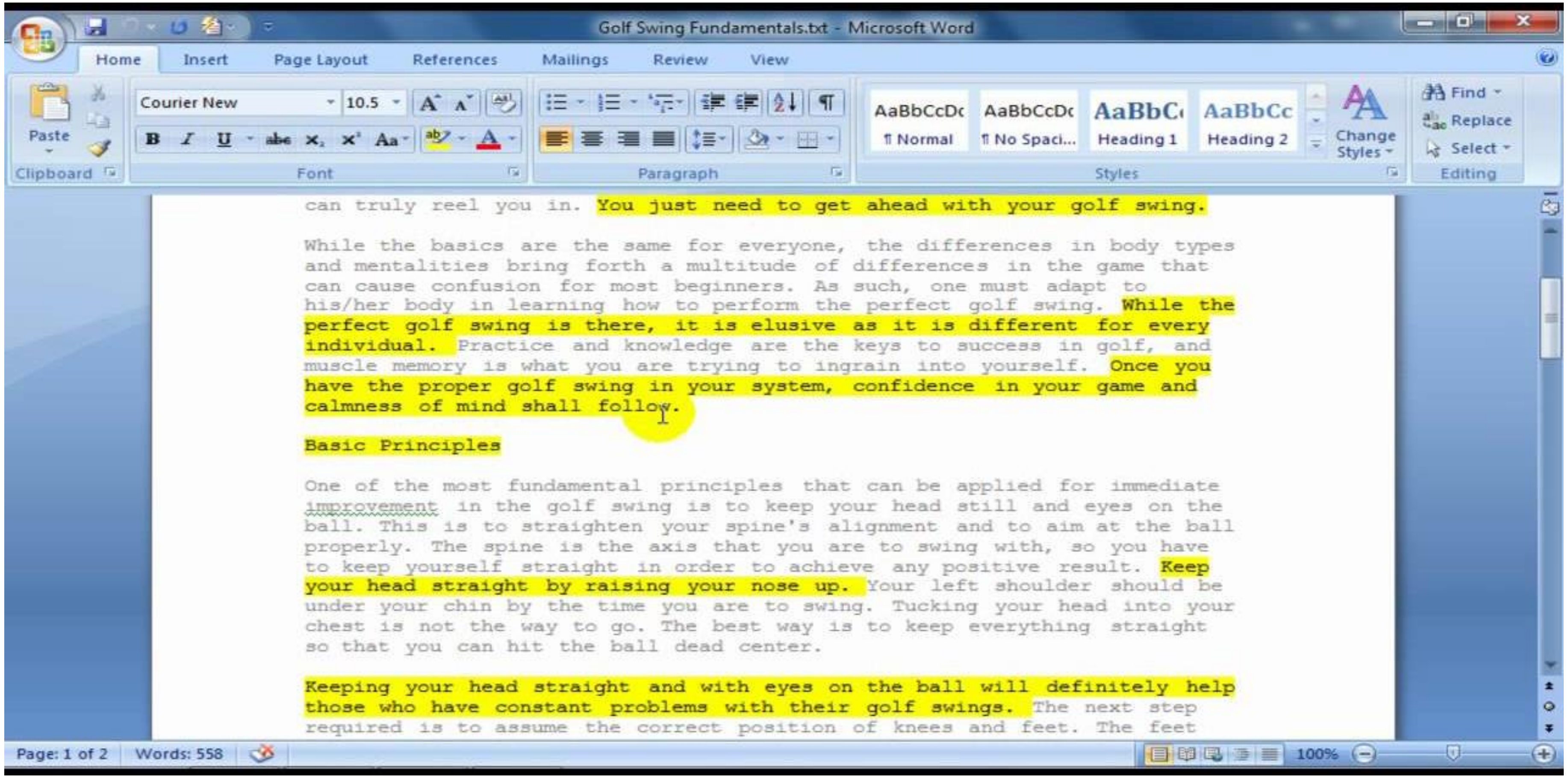
Abstract—**Adult image detection plays an important role in Internet pornographic information detection and filtering. By analyzing the shortcomings of existing pornographic image detection algorithms depending only on image content or keywords of text, a new adult image detection algorithm fusing**

describes the features of the segmentation areas with the knowledge of color, shape, texture and other characteristics [3]. But its average false rate can reach 14.1%. Especially, images of Bikini have an even higher false reject rate. Thoms et al. proposed bag-of-visual-words model for image feature

Question Answering System



Text Summarization



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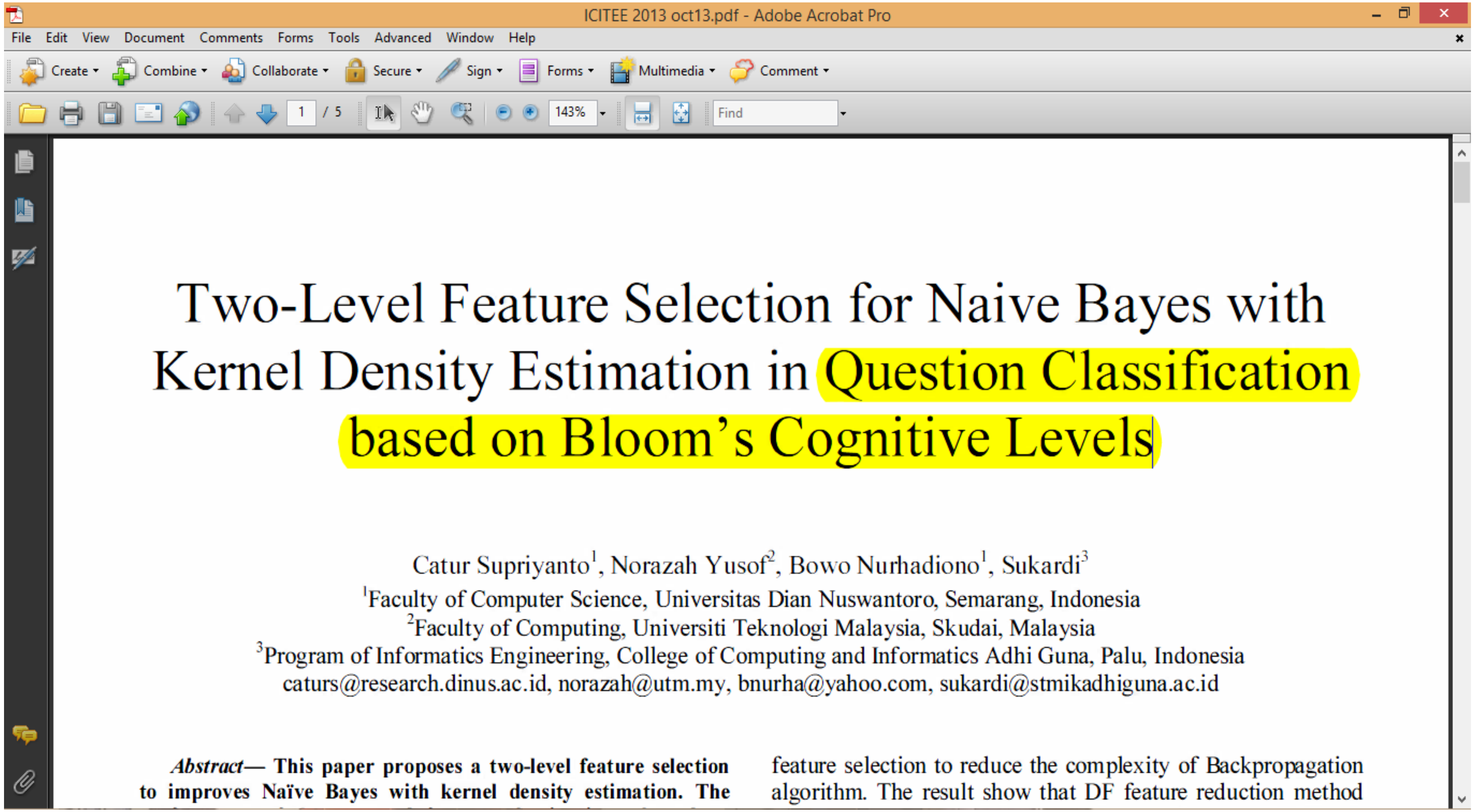

Two-Level Feature Selection for Naive Bayes with Kernel Density Estimation in Question Classification based on Bloom's Cognitive Levels

Catur Supriyanto[1], Norazah Yusof[2], Bowo Nurhadiono[1], Sukardi[3]
[1]Faculty of Computer Science, Universitas Dian Nuswantoro, Semarang, Indonesia
[2]Faculty of Computing, Universiti Teknologi Malaysia, Skudai, Malaysia
[3]Program of Informatics Engineering, College of Computing and Informatics Adhi Guna, Palu, Indonesia
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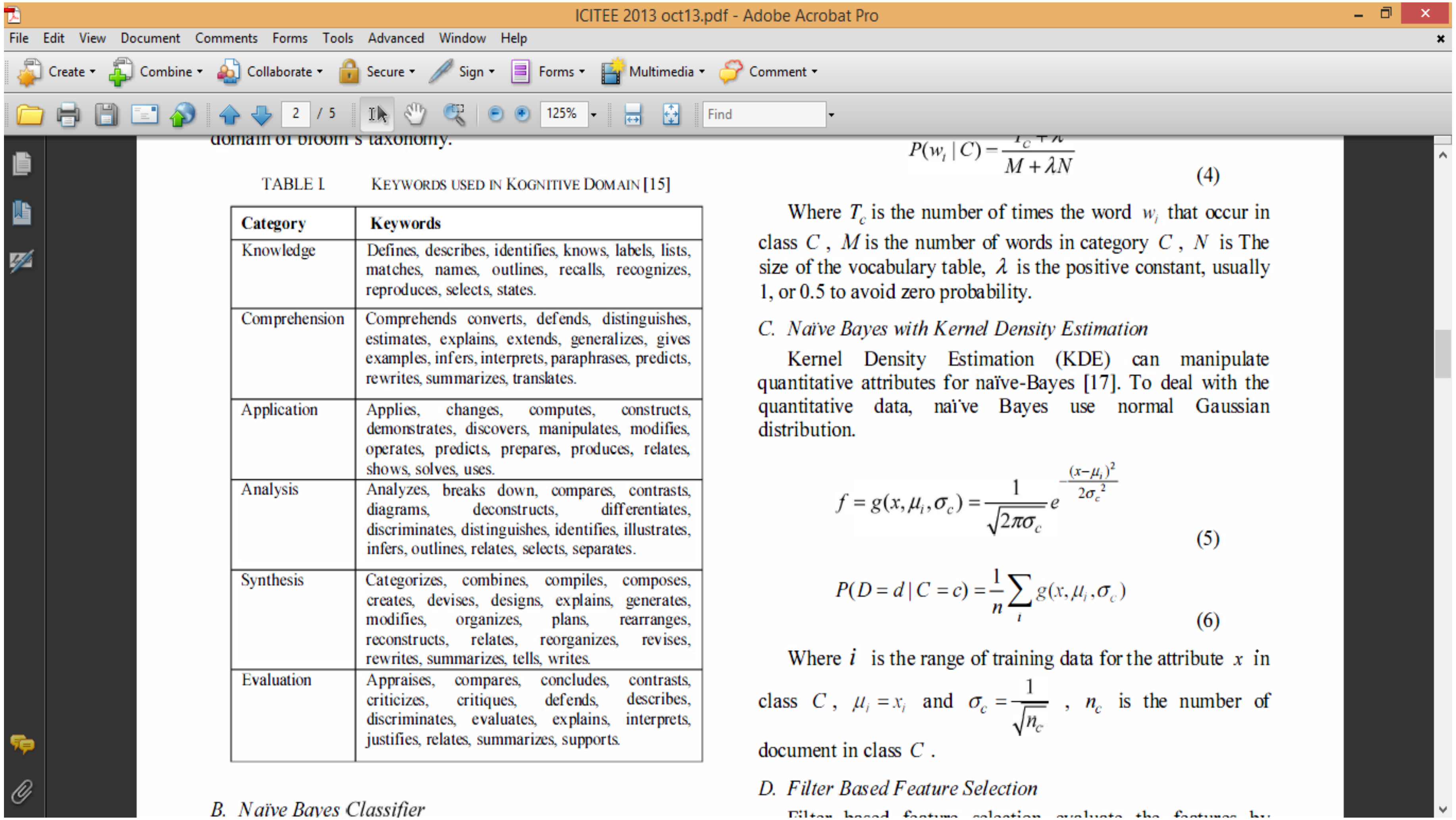
***Abstract*— This paper proposes a two-level feature selection to improves Naïve Bayes with kernel density estimation. The**

feature selection to reduce the complexity of Backpropagation algorithm. The result show that DF feature reduction method

Bloom Taxonomy



domain of bloom's taxonomy.

TABLE I. KEYWORDS USED IN KOGNITIVE DOMAIN [15]

Category	Keywords
Knowledge	Defines, describes, identifies, knows, labels, lists, matches, names, outlines, recalls, recognizes, reproduces, selects, states.
Comprehension	Comprehends converts, defends, distinguishes, estimates, explains, extends, generalizes, gives examples, infers, interprets, paraphrases, predicts, rewrites, summarizes, translates.
Application	Applies, changes, computes, constructs, demonstrates, discovers, manipulates, modifies, operates, predicts, prepares, produces, relates, shows, solves, uses.
Analysis	Analyzes, breaks down, compares, contrasts, diagrams, deconstructs, differentiates, discriminates, distinguishes, identifies, illustrates, infers, outlines, relates, selects, separates.
Synthesis	Categorizes, combines, compiles, composes, creates, devises, designs, explains, generates, modifies, organizes, plans, rearranges, reconstructs, relates, reorganizes, revises, rewrites, summarizes, tells, writes.
Evaluation	Appraises, compares, concludes, contrasts, criticizes, critiques, defends, describes, discriminates, evaluates, explains, interprets, justifies, relates, summarizes, supports.

B. Naïve Bayes Classifier

$$P(w_i \mid C) = \frac{T_c + \lambda}{M + \lambda N} \quad (4)$$

Where T_c is the number of times the word w_i that occur in class C, M is the number of words in category C, N is The size of the vocabulary table, λ is the positive constant, usually 1, or 0.5 to avoid zero probability.

C. Naïve Bayes with Kernel Density Estimation

Kernel Density Estimation (KDE) can manipulate quantitative attributes for naïve-Bayes [17]. To deal with the quantitative data, naïve Bayes use normal Gaussian distribution.

$$f = g(x, \mu_i, \sigma_c) = \frac{1}{\sqrt{2\pi}\sigma_c} e^{-\frac{(x-\mu_i)^2}{2\sigma_c^2}} \quad (5)$$

$$P(D = d \mid C = c) = \frac{1}{n} \sum_i g(x, \mu_i, \sigma_c) \quad (6)$$

Where i is the range of training data for the attribute x in class C, $\mu_i = x_i$ and $\sigma_c = \frac{1}{\sqrt{n_c}}$, n_c is the number of document in class C.

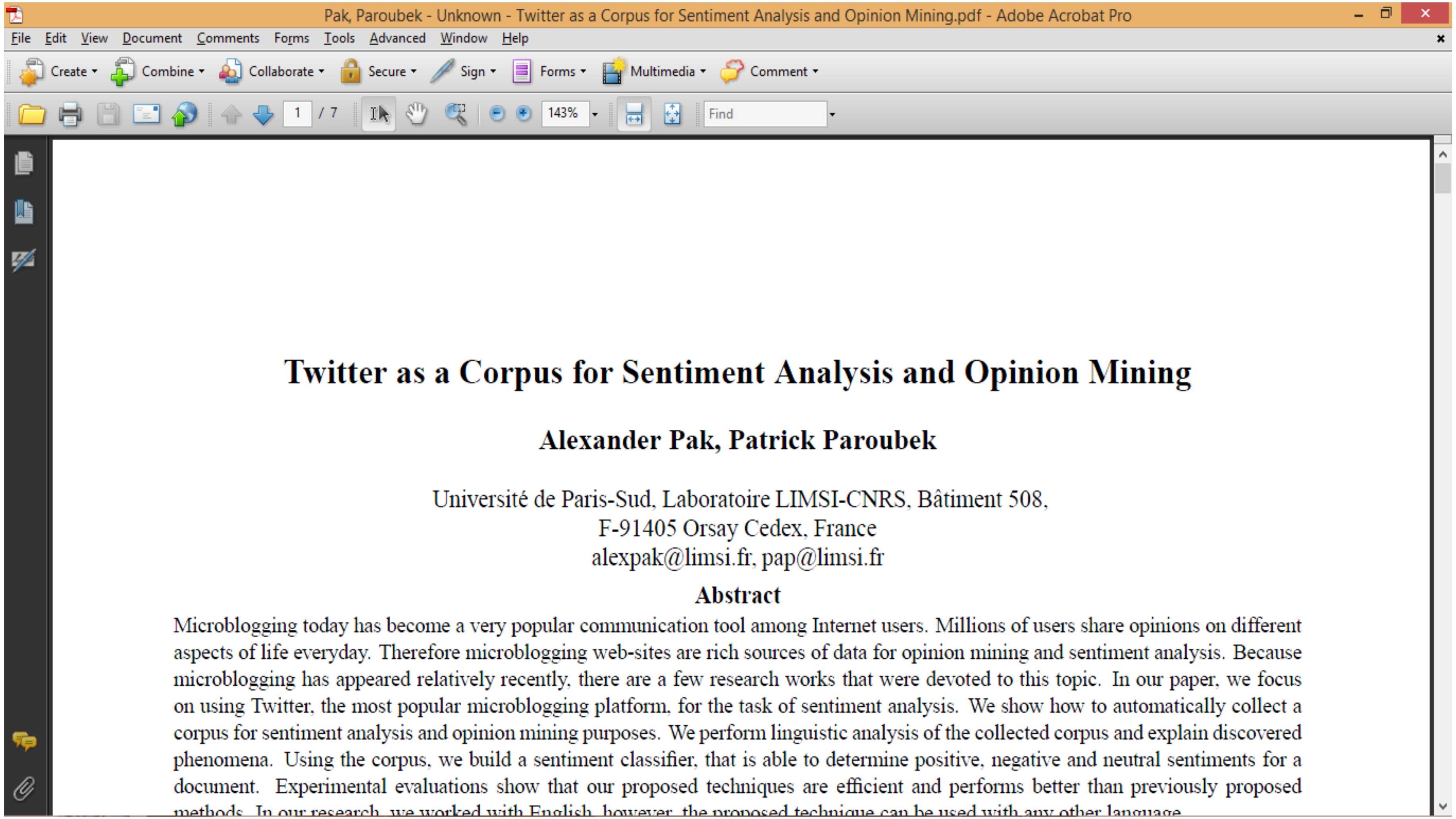
D. Filter Based Feature Selection

Filter based feature selection evaluate the features by

Question Classification Dataset

No	Questions	Label
1	Define the meaning of the Olympic Motto.	Knowledge
2	Identify the correct definition of osmosis.	Knowledge
3	Identify the standard peripheral components of a computer.	Knowledge
4	List the characteristics peculiar to the Cubist movement.	Knowledge
5	List the levels in Bloom's Taxonomy.	Knowledge
6	List the steps involved in titration.	Knowledge
7	List three characteristics that are unique to the Cubist movement.	Knowledge
8	Make a list of facts you learned from the story.	Knowledge
9	Name the food groups and at least two items of food in each group.	Knowledge
10	Name the four major food groups.	Knowledge
11	Name the main characters in the story.	Knowledge
12	Quote what ICT stands for.	Knowledge
13	Recite a policy.	Knowledge
14	State the pattern for expressing time in French.	Knowledge
15	State the rule for balls and strikes in baseball.	Knowledge
16	State the rule for using a semicolon in a sentence.	Knowledge
17	Define compound interest.	Knowledge
18	Define mercantilism.	Knowledge
19	Define stream bank, floodplain and substrate.	Knowledge
20	Draw and label a diagram of a typical stream.	Knowledge

Sentiment Analysis



Twitter as a Corpus for Sentiment Analysis and Opinion Mining

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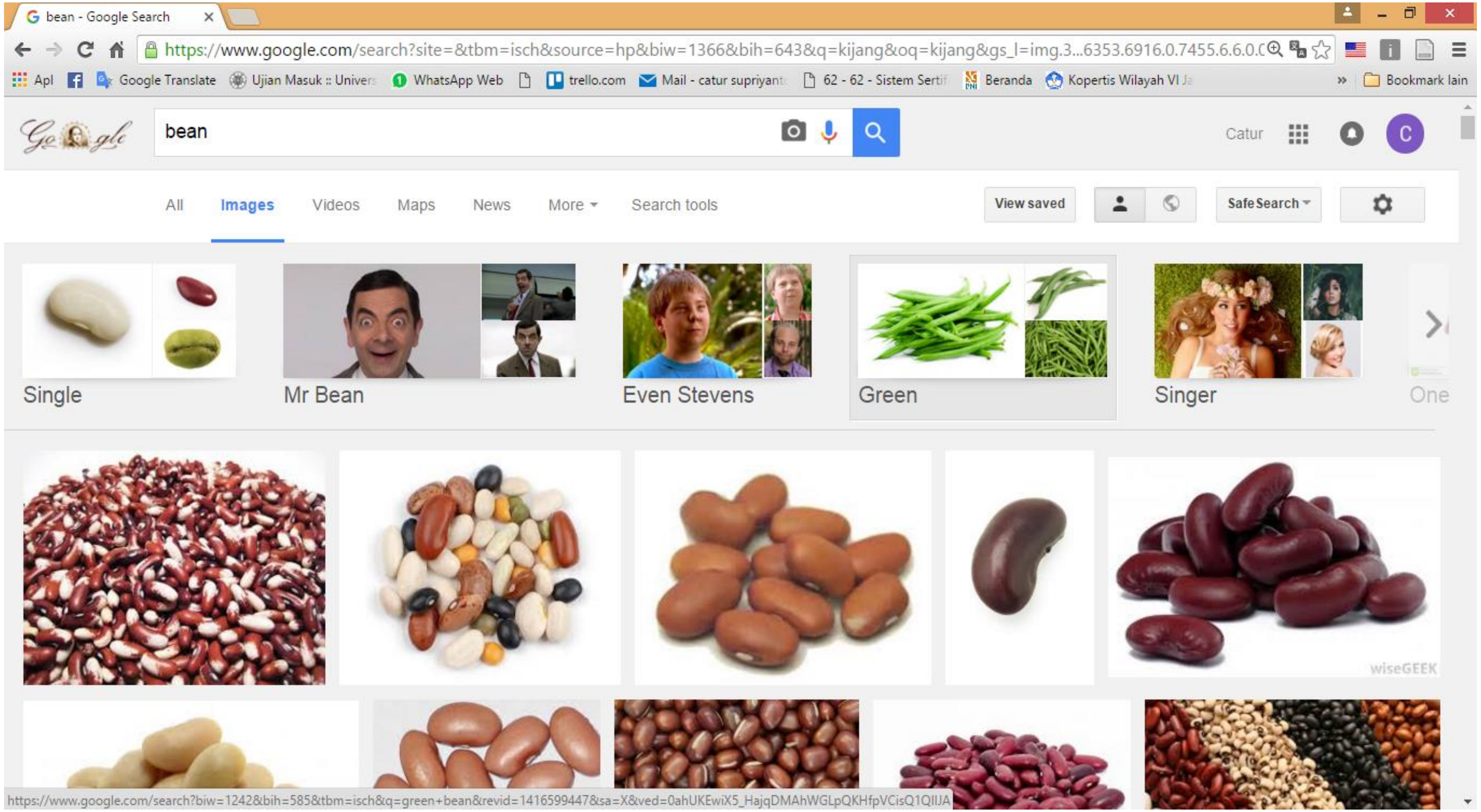
Abstract

Microblogging today has become a very popular communication tool among Internet users. Millions of users share opinions on different aspects of life everyday. Therefore microblogging web-sites are rich sources of data for opinion mining and sentiment analysis. Because microblogging has appeared relatively recently, there are a few research works that were devoted to this topic. In our paper, we focus on using Twitter, the most popular microblogging platform, for the task of sentiment analysis. We show how to automatically collect a corpus for sentiment analysis and opinion mining purposes. We perform linguistic analysis of the collected corpus and explain discovered phenomena. Using the corpus, we build a sentiment classifier, that is able to determine positive, negative and neutral sentiments for a document. Experimental evaluations show that our proposed techniques are efficient and performs better than previously proposed methods. In our research, we worked with English, however, the proposed technique can be used with any other language.

Image Clustering



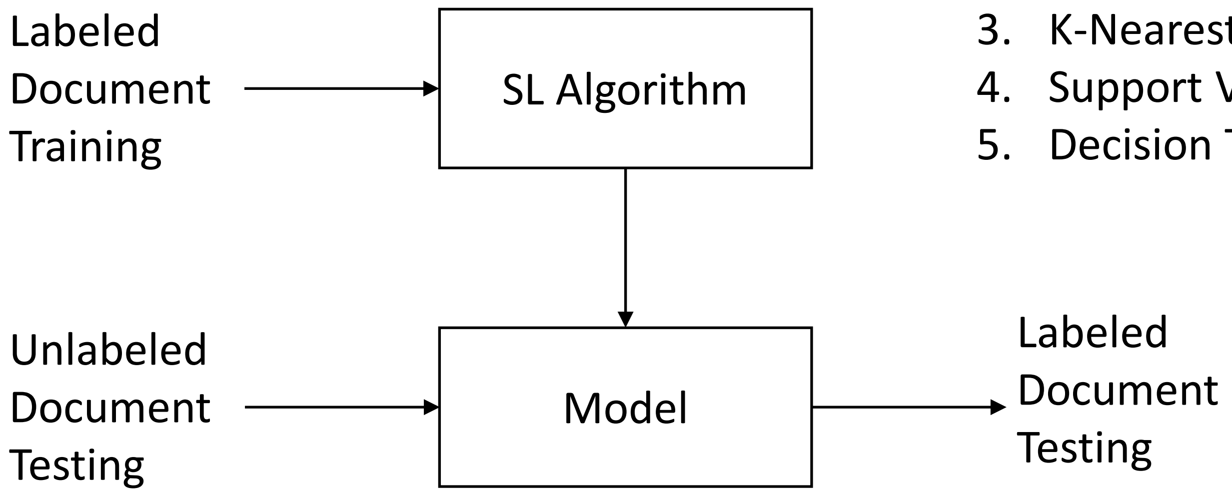
Machine Learning for Text Mining

- Text mining typically apply machine learning techniques, such as supervised learning and unsupervised learning.
- Supervised learning is referred to classification.
- Unsupervised learning is referred to clustering.

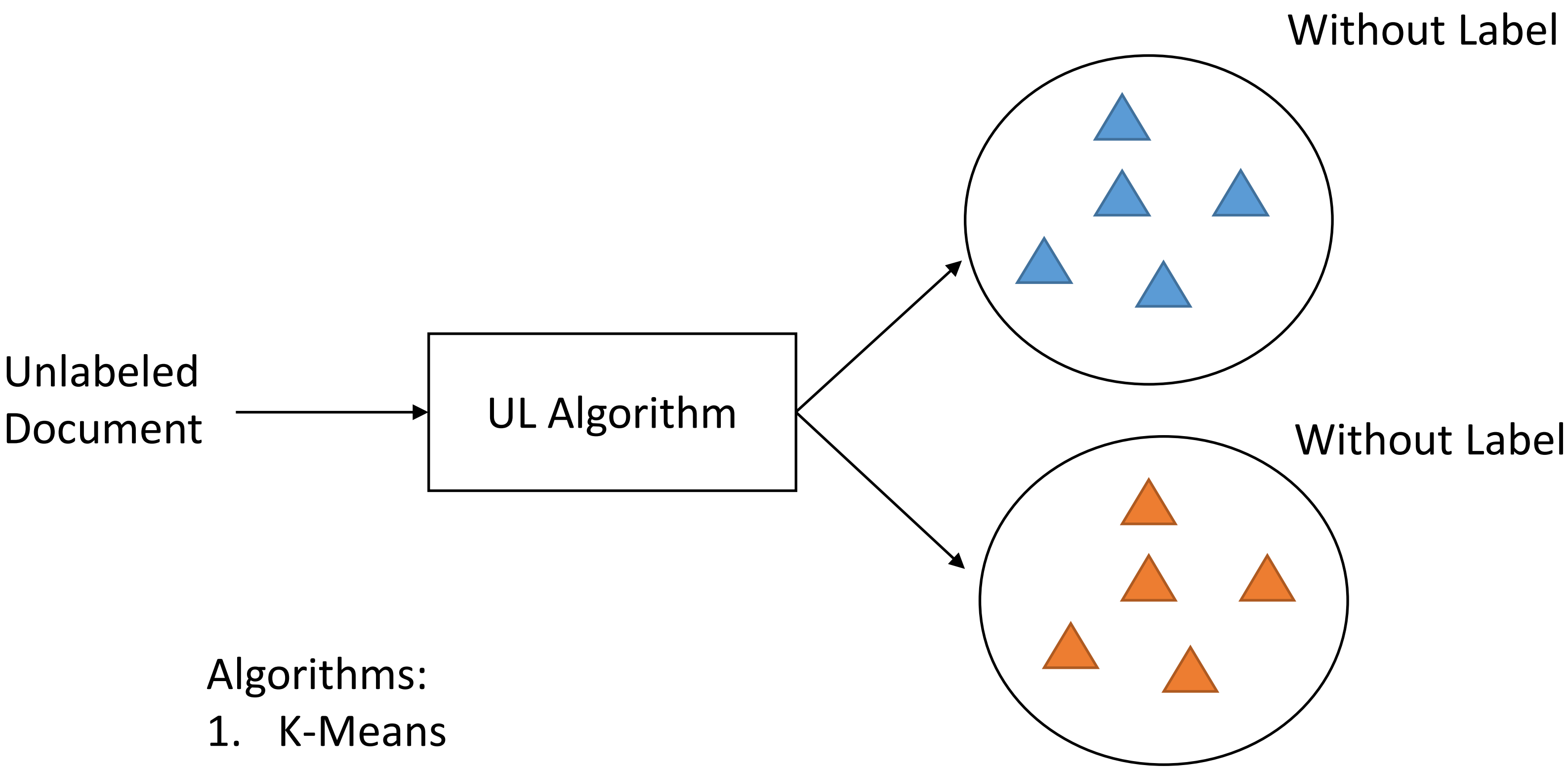
Supervised Learning (SL)

Label:
1. Sport
2. Economy
3. Politic
4. Health

Algorithms:
1. Neural Network
2. Naïve Bayes
3. K-Nearest Neighbor
4. Support Vector Machine
5. Decision Tree



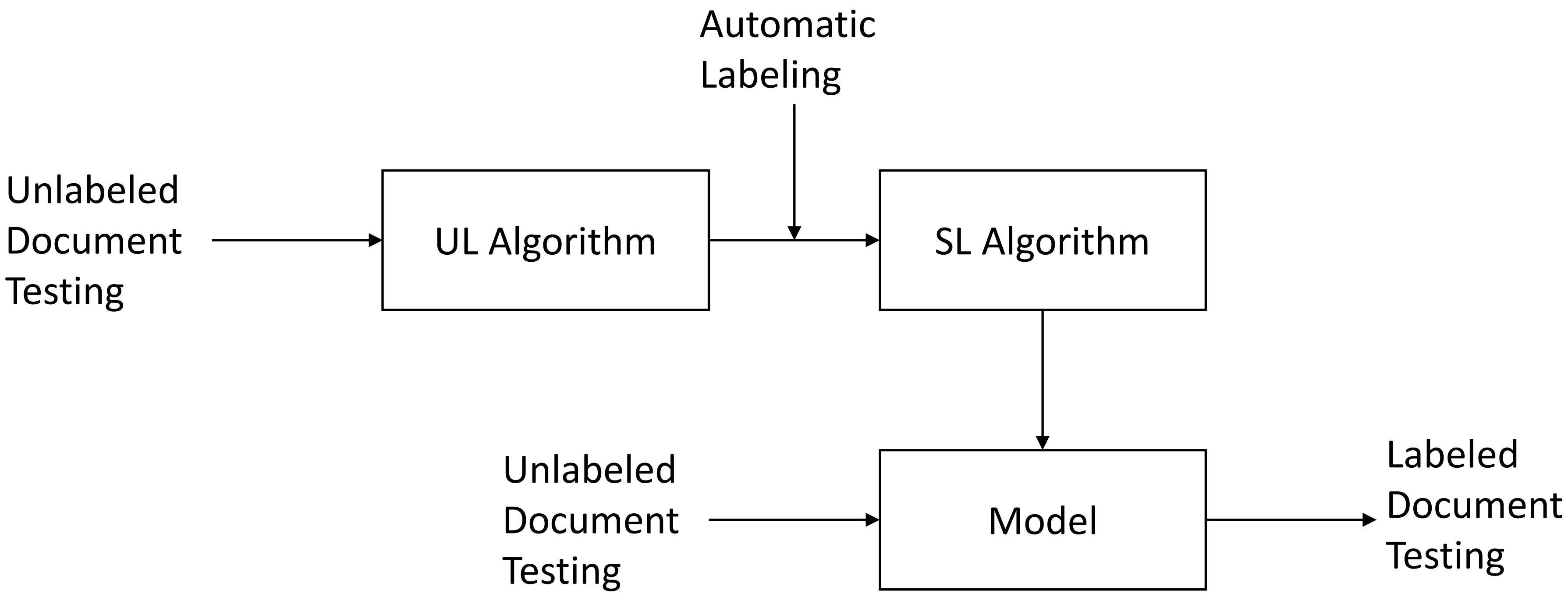
Unsupervised Learning (UL)



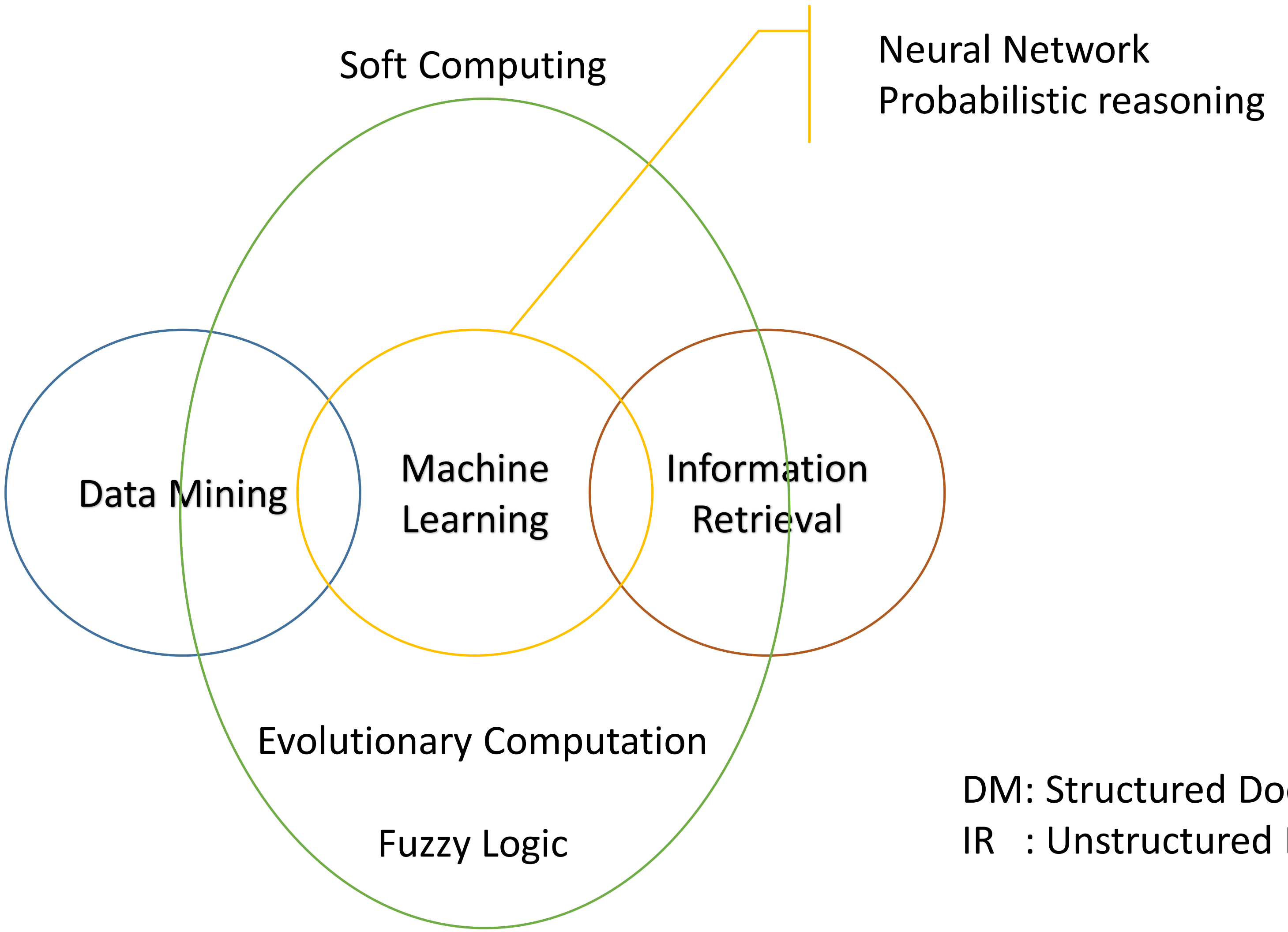
Algorithms:

1. K-Means
2. K-Medoid
3. Hierarchical Agglomerative Clustering (HAC)
4. Fuzzy C-Means (FCM)
5. Self Organizing Map (SOM)

Supervised Learning (SL) and Unsupervised Learning (UL)

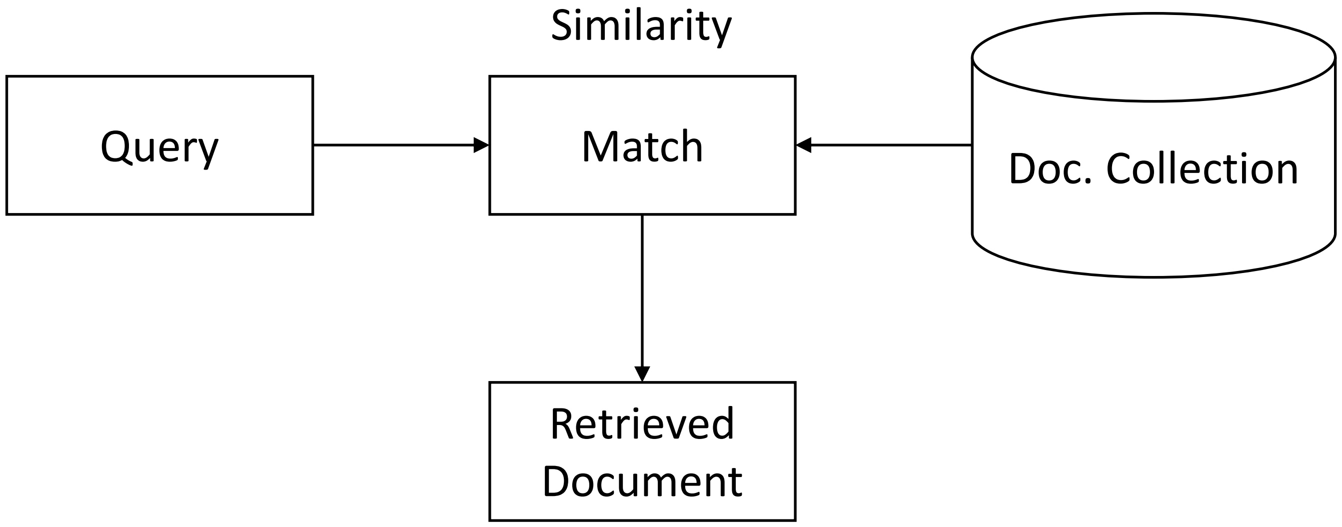


DM vs IR



DM: Structured Document
IR : Unstructured Document

Searching


Similarity
Query
Match
Doc. Collection
Retrieved
Document

Cosines Similarity

$$Sim(x, y) = \frac{\sum_{i=1}^{N} x_i \times y_i}{\sqrt{\sum_i^N (x_i)^2} \times \sqrt{\sum_i^N (y_i)^2}}$$